Chloaeilius

The spatial audio version of Chloaelis: an Ambisonics encoder (1st to 7th order), its echoes flying round you on the flock and the Lorenz attractor.

Manual for version 0.3.0



About this manual

This manual covers Chloaeilius 0.3.0, the plug-in (AU, VST3, LV2) and the desktop app. Chloaeilius is the same instrument as Chloaelis, so after the spatial chapters it carries Chloaelis's own: what it is, the layout, the symbols and how to play it. Then a reference for every parameter, and answers to common problems.

> **Spatial audio.** Chloaeilius is the Ambisonics version (1st to 7th order, 3rd by default) of Chloaelis, kept in the Chloaelis repository (this folder). It is an **encoder**: stereo goes in, Ambisonics come out (ACN order, SN3D, the format of the IEM and SPARTA plug-ins: 16 channels at 3rd order, 36 at 5th, 64 at 7th), and it decodes nothing itself. Route all of them to a decoder of the same order. For the ordinary stereo instrument, see Chloaelis.



The manual (PDF): the sphere, the desktop app, Chloaelis's controls, every parameter and answers to common problems.

The sphere

Chloaeilius is an encoder. It sends Ambisonics (ACN order, SN3D, the format of the IEM and SPARTA plug-ins) and decodes nothing itself. The **order** switch at the top left picks how sharp it is, from 1st to 7th order: 3rd (the default) is 16 channels, 5th is 36, 7th is 64. In REAPER, give the track that many channels and put the IEM AllRADecoder (loudspeakers) or BinauralDecoder (headphones) after it at the same order, SN3D. On a track with too few channels for the order you picked, the highest order that fits comes out. At 7th order all 18 points take about 14 % of one core, against 10 % at 3rd.

Where the sound goes:

- Each side's **dry sound** is a point: the left side ahead and to the left, the right side ahead and to the right.
- Each side's **main echo** rides the Lorenz attractor (the right side's mirrored through you), so the delay's repeats loop round you in the butterfly's figure.
- Each of the **seven poly taps** per side rides one bird of the flock, now flying in three dimensions round you. A feedback trail comes back from wherever its bird has flown, so repeats drift through the room, and the birds gather and scatter round you rather than to one side.
- **Distance**, after the IEM RoomEncoder: every point has its own propagation delay (so echoes flying towards you or away bend in pitch), 1/r level, a little air absorption and first-order reflections from the walls of a room round you. The **Distance** knob (lambda) is the size of the bubble you're in, from 1.5 m to 24 m: in a small bubble everything is close and loud and the room answers quickly; in a big one everything is far, quieter, later and roomier.
- With the order on *delay > filter*, the dry sound goes through the tube and filter as before and each tap through its own filter (the taps skip the tube there, so they stay apart).

Space View, the switch of four marks at the top (∞ ⊙ ⊞ ◎), shows the sphere four ways (the sound is the same in all of them):

∞ **Lorenz**: the attractor alone, as in Chloaelis



⊙ **Orrery**: a dotted globe round you, turning slowly and leaning with the cursor; every point flies inside it with its trail, and the loud places glow on its skin



⊞ **Map**: the whole sphere unrolled (a Hammer projection, as the IEM EnergyVisualizer), ahead in the middle, behind at both edges, with the energy as glowing dust



◎ **Radar**: from above, rings in metres out to the bubble's edge, each point on a stalk as tall as it is high, and a ripple wherever an echo lands



The sphere button (the circle under **?** in the top right) opens a big sphere over the whole window, with nothing in the way: drag to turn it, scroll to zoom, double-click to set it straight, and switch between globe (⊙), map (⊞) and radar (◎) along the top. Where things are is marked by which half of a disc is lit: ◓ ahead, ◐ left, ◑ right, ◒ behind. Every point is named (L DRY, L ECHO, L1 to L7 for the taps, and the same for R), and the loud places glow on the glass.



The desktop app

`Chloaeilius.app` (`Chloaeilius.exe` on Windows, `Chloaeilius` on Linux) is Chloaeilius on its own, with no DAW, and it has Chloaelis's desktop extras:

- **The vessel** (top right): load your own sample. Drop an audio file onto it (WAV, AIFF, FLAC, MP3, M4A, OGG or CAF, up to 10 minutes) or double-click it to choose one. The two rings set the loop, and dragging the glass sets its level. It plays into both sides and its echoes fly round you like everything else.
- **Listen** (top left), because without a DAW there's nowhere to put a decoder:
  - ∩ **Headphones**: the sphere decoded to binaural (a spherical-head model), on outputs 1 and 2.
  - ⟨⟩ **Speakers**: a stereo decode for two loudspeakers.
  - ⁙ **16 ch**: the Ambisonics themselves, for a multichannel interface or to route into another app (for example through BlackHole into REAPER and an IEM decoder).
- **Options** (the audio device and channels), **fullscreen** (cmd+F, esc to leave) and the help, as in Chloaelis.

The plug-ins (AU, VST3, LV2) have no Listen switch: in a DAW they always send the Ambisonics to your decoder. The desktop app works at 3rd order.



Build it

Ready-made builds for macOS, Windows and Linux are on the Chloaelis Releases page, tagged `chloaeilius-v...` (separate from Chloaelis's own). To build it yourself, use this folder (it shares the repository's JUCE, fonts and recordings, so clone the whole thing):

```
git clone --recursive https://github.com/femifleming/Chloaelis.git
cd Chloaelis
cmake -S Chloaeilius -B build-chloaeilius -DCMAKE_BUILD_TYPE=Release -
DCHLOAEILIUS_COPY_AFTER_BUILD=ON
cmake --build build-chloaeilius --config Release --target Chloaeilius_All
```

It appears in your DAW as **Chloaeilius** under **Chloaelis**. The tests are `ChloaeiliusTest`, `ChloaeiliusSpatialTest` (where the sound lands, every channel against AmbiX, CPU) and `ChloaeiliusHostTest` (`SPACE_SHOTS=<folder>` saves every view of the sphere).

Everything else works as in Chloaelis, including what came with its version 1.1: Karplus-Strong time (ϙ) tuned to Cloudfin's scales, the tape hiss that fades instead of cutting, symbols in place of every word, and the Brick look (▦). In the sphere, a string's poly taps still each ride their own bird, so a chord is spread round you. See its README.

Human contributions

Chloaeilius was written with Claude Code (Anthropic) under the direction of its author, femifleming. It is the spatial version of Chloaelis. The idea, the references (Bizarre Jezabel's *Chloe* and the game *Splice*), the look and the field recordings all come from the author's work on Chloaelis: see Human contributions in its README.

What it is (as in Chloaelis)

Chloaelis is two complete signal paths, hard-panned: the **left input goes through the left side to the left output**, and the right through the right. Each side runs

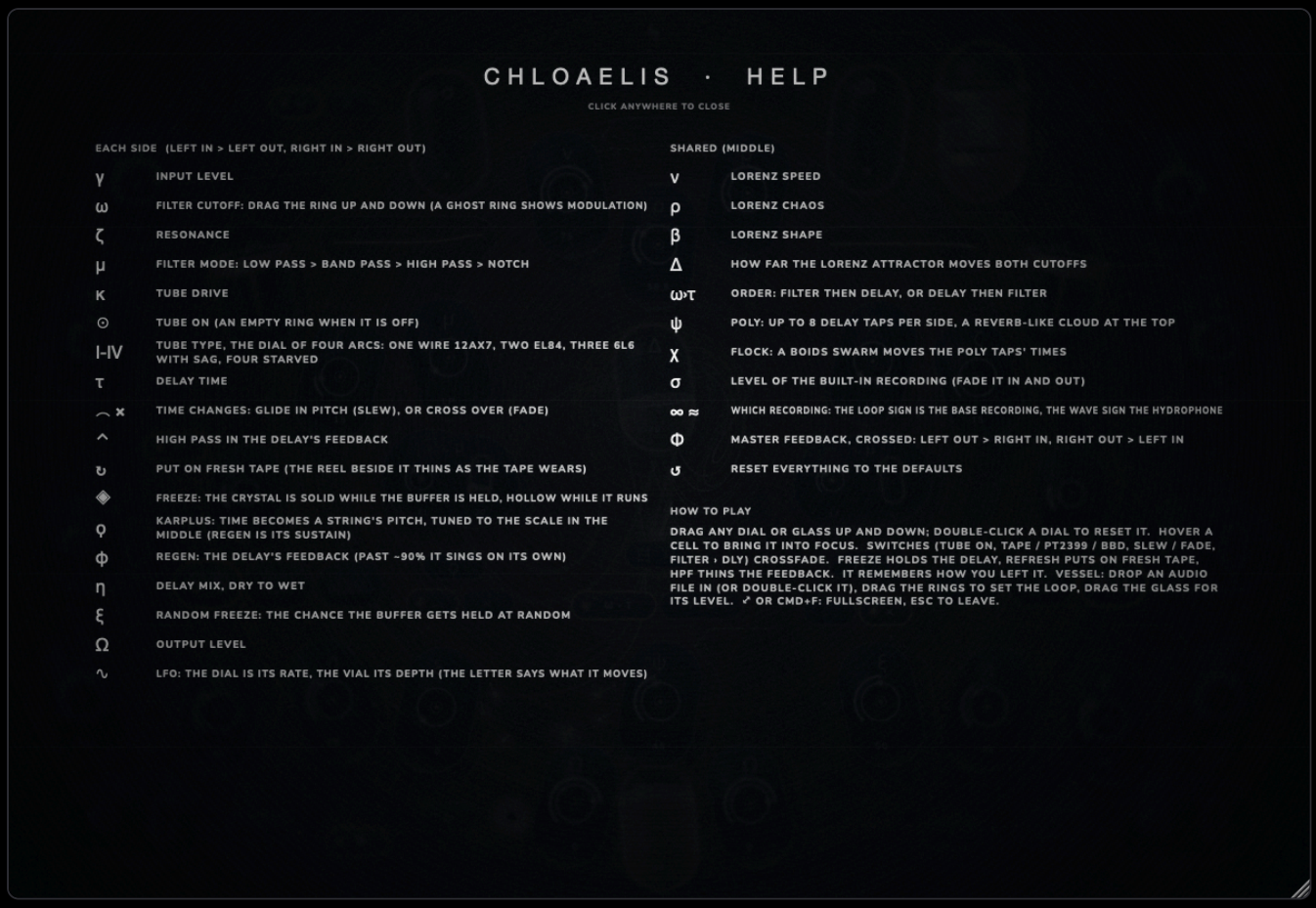
input > multimode filter > tube > delay > output

(or delay > tube > filter, with the order switch). Shared controls in the middle tie the two sides together:

- **Filter**: continuous low pass > band pass > high pass > notch, with resonance.
- **Tube**: four types, each oversampled 4x: I 12AX7 (the default: warm, two triode stages), II EL84 (push-pull pentodes, crunchy), III 6L6 (with power-supply sag, it blooms) and IV starved (a low-voltage 12AX7, gated and sputtery).
- **Delay engines**: *tape* (an ageing tape loop: hiss, wow and flutter, oxide wear; it ages while it runs and *refresh* puts on new tape), *PT2399* (the delay chip, gritty at long times) and *BBD* (bucket brigade, dark and crunchy). Up to 1 s, regen that self-oscillates, a feedback high-pass, freeze, random freeze, and slew (pitch-bending) or fade time changes. The tape's hiss never cuts in or out: it fades up when you switch to tape and down when you leave it, so a recording has no step.
- **Karplus-Strong time** (ϙ): the delay becomes a tuned string. The time knob picks its note, snapped to one of 25 scales (the list from Cloudfin: La Monte Young's Well-Tuned Piano, ragas, maqamat, gamelan, Partch, Bohlen-Pierce, equal divisions), and regen is its sustain.
- **Poly**: fades in up to 8 delay taps per side at irregular offsets. All the way up it becomes a reverb-like cloud. **Flock** lets a boids simulation move those taps' times, so they drift, cluster and scatter.
- **Lorenz attractor**: its x and y move the left and right cutoffs.
- **LFOs**: triangle LFOs on filter mode, delay time and tube drive, per side.
- **Master feedback**, crossed over: the left output feeds the right input and the right output the left.
- **Two built-in recordings** to play through it: a long loop from *baserecording1* and a hydrophone recording. Fade them in and out; the left channel goes to the left side and the right to the right.
- Every switch (order, delay engine, time mode, tube type) **crossfades**: echoes ring out in the old setting while new sound goes through the new one, so there are no cuts.
- It **remembers** how you left it, and ↺ resets everything.
- **No words on its face**: every control is a letter, a mark or a picture (the **?** screen is the key). And two looks: the glass HUD, or **Brick** (▦), the Windows 2000 colour scheme in hard pixels.

How to use it (as in Chloaelis)

The layout



- The **left half** of the window is the left channel and the **right half** is the right channel. They mirror each other: input at the top, then filter, tube and LFOs, delay at the bottom, output at the bottom middle.
- The **middle** holds the shared controls: the Lorenz attractor (top), the poly and flock (bottom), the order switch, and the built-in recordings, master feedback and the vessel (top row).
- **Glowing dashed lines** show where signal and modulation flow. Pulses of light follow the signal.
- Cells **glow with level**: the side's colour when it's moving, warm white when it's hot and red when it's close to clipping.
- The **Lorenz attractor** floats in the background as a hologram, with dashed lines to the two cutoffs. The **boids** of the flock fly around the poly controls.

The symbols

There are no words on the face: every control is a Greek letter, a mark or a small picture. Press **?** (top right) for this list inside the app.

Each side

Symbol	Control	What it does
γ	Input	input level
ω	Cutoff	filter cutoff: drag the ring up and down. A faint ring shows where modulation has pushed it

ζ	Resonance	filter resonance
μ	Mode	filter mode: low pass > band pass > high pass > notch, continuously
κ	Drive	tube drive
the dial of four arcs	Tube type	the lit arc and the wires in its lens count the type: one 12AX7 (default), two EL84, three 6L6 with sag, four starved. Click for the next (alt- or right-click for the one before), or scroll
⊙ / ○	Tube on	switches the tube in and out: a lit heater, or an empty ring
τ	Time	delay time, 30 ms to 1 s
φ	Regen	the delay's feedback. Past about 90% it sings on its own
η	Mix	delay dry / wet
ξ	Random freeze	the chance that the delay buffer gets held for a while at random moments
reels / chip / buckets	Delay engine	three pictures: two reels (tape), a chip with bits going through it (PT2399), a row of buckets passing charge along (BBD). The one in use moves
⌒ / ×	Time changes	either glide (slew: pitch bends) or cross over (fade: no pitch change)
ϙ	Karplus-Strong time	the delay becomes a tuned string: see below
^	High pass	high-pass filter in the delay feedback
◈ / ◇	Freeze	holds the delay buffer. The crystal is solid while it is held (by hand or by the random freeze), hollow while it runs. On tape, a held loop slowly wears away
ʊ and the reel	Fresh tape	puts on fresh tape (tape engine only). The reel beside it is the tape's condition: its pack is fat when new and thins as the tape wears
Ω	Out	output level
∿	LFO	the dial is its rate, the small glass vial its depth. The letter next to ∿ says what it moves (μ mode, τ time, κ drive)

Shared (middle)

Symbol	Control	What it does
ν	Speed	how fast the Lorenz attractor moves
ρ	Chaos	the attractor's rho: low wanders gently, high is wild
β	Shape	the attractor's beta: reshapes the orbit
Δ	Depth	how far the attractor moves the two cutoffs (up to ±3.5 octaves)
ω › τ / τ › ω	Order	filter > tube > delay, or delay > tube > filter (the filter's and the delay's own letters, in order)
the ladder and a number	Scale	the scale the Karplus-Strong notes are tuned to, and how many notes it has. Hover to read its name, click for the list, or scroll through it
ψ	Poly	up to 8 delay taps per side. All the way up is a reverb-like cloud

χ	Flock	how much the boids move the poly taps' times (up to about ±0.9 octave)
∞ / ≈	Source	which built-in recording: ∞ the base loop, ≈ the hydrophone
σ	Source level	fades the built-in recording in and out
Φ	Master feedback	crossed: left out > right in, right out > left in
↺	Reset	every control back to its default (the tape's wear is kept)
▦	Brick	switches the look (top left): the glass HUD, or Brick
?	Help	the key to all of this, inside the app

Karplus-Strong time

ϙ (koppa, the old Greek K) on either side turns that side's delay into a Karplus-Strong string: a very short, exactly tuned delay that rings at a pitch.

- The **time knob (τ)** now picks a **note**. It still runs the same way round (up is longer, so lower): six octaves, from 27.5 Hz at the top of the knob to 1760 Hz at the bottom, and its readout shows the pitch in Hz. Notes snap to the chosen scale, whose 1/1 is A.
- The **scale** is the small cell in the middle (a ladder and the number of notes in the scale). Hover over it to read the name, click it for the list, or scroll. They are the scales of Cloudfin: The Well-Tuned Piano (La Monte Young), twelve ragas, 22 shruti, three maqamat, slendro and pelog, the harmonic series, Partch's 43 tones, Bohlen-Pierce, and 24-, 19-, 31- and 12-EDO.
- **Regen (φ)** is the string's sustain: most of its travel is long rings, and past 100% it sings on its own. **Freeze** holds the string.
- **Poly (ψ)** adds strings on other degrees of the scale, stacked up from the note you are on, so it plays a chord. **Flock (χ)** only detunes them a little.
- The **engine** pictures set the string's colour here: PT2399 bright, tape softer, BBD dark.
- **Slew (⌒)** glides between notes and **fade (×)** steps to them. The time LFO walks through the scale.

The two looks

▦ (top left) switches between the glass HUD and **Brick**: the Windows 2000 "Brick" colour scheme, with bevelled capsules, sunk white dials, progress-bar blocks in the vials and a maroon graph for the filter, all in hard pixels with no blur, glow or drift. The choice is kept with the state.

Playing it

- **Drag up and down** on any dial, glass vial or ring to change it. **Double-click** a dial to reset it to its default.
- **Hover** over a control to bring it into focus. The depth of field follows you, and the field tilts slightly toward the cursor.
- **No input?** Turn up **σ** to play one of the built-in recordings through it, and pick which with **∞ / ≈**.
- **Ideas to start with**
  - *Ambient wash*: σ up, ψ (poly) and χ (flock) up, φ (regen) around 60%, tape engine.
  - *Lo-fi echo*: PT2399 or BBD, τ around 400 ms, φ around 70%, κ (drive) up a little.
  - *Moving filter*: Δ (Lorenz depth) up, ρ (chaos) up, ζ (resonance) up.
  - *Self-playing*: Φ (master feedback) towards 90% with some regen: the two sides feed each other and keep going on their own.

    - *Tuned strings*: ϙ on, φ (regen) around 90%, ψ (poly) up for a chord, and move τ or its LFO to play the scale.

- Chloaelis **remembers its last state**: a new instance starts how the last one was closed. In a DAW, the project's own saved settings take priority.

The desktop app

The app (`Chloaelis.app`, `Chloaelis.exe` or `Chloaelis` on Linux) has a few extras:

- **Audio settings**: the **Options** button at the top left of the window chooses the audio device, the input and output channels and the buffer size.
- **Fullscreen**: the ⛶ button, or **⌘F** (Ctrl+F on Windows and Linux). It fills the whole screen with no menu bar or title bar, at any screen shape. **Esc** leaves it.
- **The vessel** (top right): a glass you can pour your own audio into.
    - **Drop an audio file** onto it (WAV, AIFF, FLAC, MP3, M4A, OGG or CAF, up to 10 minutes), or **double-click** it to choose one.
    - Its waveform runs up the inside, start at the bottom.
    - The **two rings** are the loop start and end: drag them.
    - **Drag the glass** itself (away from the rings) to set its level, shown by the liquid.
    - It plays into both sides like the built-in recordings, and is remembered with the rest of the state.

Help inside the app

Press **?** (top right) for a full key to the symbols and a short how-to. Click anywhere, or press Esc in the app, to close it.

Parameter reference

Every control your DAW can automate, by the name it shows there. The two sides have the same controls, so the right side's table follows the left's. Defaults are what a fresh instance starts with (reset brings them back).

Shared (the middle)

Name in your DAW	ID	Range	Default
Lorenz Speed	`lz_speed`	0% to 100%	35%
Lorenz Chaos	`lz_chaos`	ρ 24.5 to ρ 180.0	ρ 38.5
Lorenz Shape	`lz_shape`	β 0.50 to β 4.00	β 2.67
Lorenz > Cutoff	`lz_depth`	0% to 100%	0%
Poly Width	`width`	OFF to 100%	OFF
Poly Flock	`poly_flock`	0% to 100%	0%
Order	`order`	Filter > Delay, Delay > Filter	Filter > Delay
Source	`src_sel`	Base Loop, Hydrophone	Base Loop
Source Level	`src_level`	0% to 100%	0%
Master Feedback	`master_fb`	0% to 100%	0%
Karplus-Strong Scale	`ks_scale`	The Well-Tuned Piano (La Monte Young), Raga Bhairav, Raga Yaman, Raga Bhairavi, Raga Todi, Raga Marwa, Raga Purvi, Raga Kafi, Raga Malkauns, Raga Darbari Kanada, Raga Bhupali, Raga Desh, Raga Puriya Dhanashri, 22 Shruti, Maqam Rast, Maqam Bayati, Maqam Saba, Slendro, Pelog, Harmonic Series 8-16, Partch 43-Tone, Bohlen-Pierce (tritave), Quarter-tone (24-EDO), 19-EDO, 31-EDO, Chromatic (12-EDO)	The Well-Tuned Piano (La Monte Young)
Distance	`space`	1.5 m to 24.0 m	7.9 m
Listen (desktop app)	`listen`	Headphones, Speakers, 16 ch	Headphones
Space View	`space_view`	Lorenz, Orrery, Map, Radar	Orrery
Ambisonic Order	`ambi_order`	1st, 2nd, 3rd, 4th, 5th, 6th, 7th	3rd

Left side

Name in your DAW	ID	Range	Default
L In Att	`l_inatt`	silent to 12.0 dB	0.0 dB
L Cutoff	`l_cutoff`	20 Hz to 20.0 kHz	20.0 kHz
L Resonance	`l_res`	0% to 100%	15%
L Mode	`l_morph`	LP to Notch	LP

L Drive	l_drive	0% to 100%	30%
L Time	l_time	30 ms to 1000 ms	330 ms
L Regen	l_regen	0% to 115%	40%
L Wet	l_wet	0% to 100%	35%
L Out	l_out	silent to 12.0 dB	0.0 dB
L Random Freeze	l_randfrz	0% to 100%	0%
L LFO>Mode Rate	l_lfomode_rate	0.02 Hz to 20.0 Hz	0.21 Hz
L LFO>Mode Depth	l_lfomode_depth	0% to 100%	0%
L LFO>Time Rate	l_lfotime_rate	0.02 Hz to 20.0 Hz	0.11 Hz
L LFO>Time Depth	l_lfotime_depth	0% to 100%	0%
L LFO>Drive Rate	l_lfodrive_rate	0.02 Hz to 20.0 Hz	0.37 Hz
L LFO>Drive Depth	l_lfodrive_depth	0% to 100%	0%
L Delay HPF	l_hpf	off / on	Off
L Freeze	l_freeze	off / on	Off
L Tube On	l_tubeon	off / on	On
L Delay Type	l_dtype	Tape, PT2399, BBD	PT2399
L Time Change	l_tmode	Slew, Fade	Slew
L Tube Type	l_tubetype	12AX7, EL84, 6L6, Starved	12AX7
L Karplus-Strong Time	l_ks	off / on	Off

Right side

Name in your DAW	ID	Range	Default
R In Att	r_inatt	silent to 12.0 dB	0.0 dB
R Cutoff	r_cutoff	20 Hz to 20.0 kHz	20.0 kHz
R Resonance	r_res	0% to 100%	15%
R Mode	r_morph	LP to Notch	LP
R Drive	r_drive	0% to 100%	30%
R Time	r_time	30 ms to 1000 ms	410 ms
R Regen	r_regen	0% to 115%	40%
R Wet	r_wet	0% to 100%	35%
R Out	r_out	silent to 12.0 dB	0.0 dB
R Random Freeze	r_randfrz	0% to 100%	0%
R LFO>Mode Rate	r_lfomode_rate	0.02 Hz to 20.0 Hz	0.17 Hz
R LFO>Mode Depth	r_lfomode_depth	0% to 100%	0%
R LFO>Time Rate	r_lfotime_rate	0.02 Hz to 20.0 Hz	0.13 Hz
R LFO>Time Depth	r_lfotime_depth	0% to 100%	0%
R LFO>Drive Rate	r_lfodrive_rate	0.02 Hz to 20.0 Hz	0.29 Hz
R LFO>Drive Depth	r_lfodrive_depth	0% to 100%	0%
R Delay HPF	r_hpf	off / on	Off

R Freeze	`r_freeze`	off / on	Off
R Tube On	`r_tubeon`	off / on	On
R Delay Type	`r_dtype`	Tape, PT2399, BBD	PT2399
R Time Change	`r_tmode`	Slew, Fade	Slew
R Tube Type	`r_tubetype`	12AX7, EL84, 6L6, Starved	12AX7
R Karplus-Strong Time	`r_ks`	off / on	Off

Distance is the radius of the bubble you're in. Ambisonic Order sets how many channels go out (16 at 3rd order, 36 at 5th, 64 at 7th). Listen only does anything in the desktop app; the plug-ins always send the Ambisonics. Space View changes the picture, never the sound.

Problems and answers

I only hear two channels, or nothing moves round me

In a DAW the track needs as many channels as the order switch says (16 at 3rd order, 36 at 5th, 64 at 7th) and a decoder after Chloaeilius at the same order (ACN, SN3D), such as the IEM AllRADecoder or BinauralDecoder. In the desktop app, set Listen to Headphones or Speakers.

macOS says it can't be opened, or the DAW doesn't list it

The builds aren't notarised. Run `xattr -dr com.apple.quarantine` on the app and the plug-ins (see Installing), then rescan in your DAW.

There's no sound and nothing on the input

Turn up σ (source level) to play a built-in recording through it, and pick which with ∞ / ≈. In the desktop app, check Options for the audio device.

It keeps sounding after I stop

Regen past about 90 % sings on its own, and master feedback (Φ) keeps the two sides feeding each other. Turn them down, or reset with ↺.

A new instance doesn't start from the defaults

It remembers its last state, so a new instance starts how the last one was closed. A saved project still opens as you saved it. ↺ resets everything (the tape's wear is kept).